GymPulse — Technology Stack

Multi-tenant Gym Management SaaS Platform

Project Overview

GymPulse is a comprehensive multi-tenant gym management platform built as a modern SaaS application. It handles member management, attendance tracking, billing/subscriptions, retention analytics with ML-powered churn prediction, marketplace for vendors/products/services, automated data import pipelines with OCR, and financial ledger systems — all with Nepal-specific localization (NPR currency, Nepali calendar, Asia/Kathmandu timezone).

Core Framework

Technology	Version	Purpose
Next.js	16.3.6	App Router, React Server Components, Serverless Functions
React	19.2.8	UI Library (Latest concurrent features)
React DOM	19.2.8	DOM Rendering
TypeScript	5.x	Type-safe Development (Strict Mode, ES2022)

Database & ORM

Technology	Version	Purpose
PostgreSQL	Latest	Primary Relational Database (via Prisma Adapter)
Prisma ORM	7.10.0	Schema-first ORM, Type-safe Client, Migrations
@prisma/adapter-pg	7.10.0	PostgreSQL Driver Adapter for Prisma
pg	8.23.0	Raw PostgreSQL Queries

Authentication & Security

Technology	Version	Purpose
@node-rs/argon2	2.2.1	Password Hashing (Native WASM, Argon2id)

otplib	13.5.0	TOTP/MFA (RFC 6238 Compliant)
Custom Session Mgmt	—	JWT-like Tokens with DB Storage & Revocation
RBAC System	—	Role-based Access (Owner/Manager/Staff/Trainer) + Granular Permissions

UI & Styling

Technology	Version	Purpose
Tailwind CSS	4.x	Utility-first CSS (PostCSS Plugin)
Radix UI	1.6.7	Headless Accessible Component Primitives
class-variance-authority	0.7.1	Component Variant Management
clsx	2.1.1	Conditional Class Names
tailwind-merge	3.7.0	Tailwind Class Merging (Deduplication)
lucide-react	1.48.0	Icon System
motion (Framer Motion)	13.4.2	Animations & Transitions
recharts	3.10.1	Charts & Analytics Visualizations
sonner	2.0.8	Toast Notifications
cmdk	1.1.1	Command Palette / Search

Forms & Validation

Technology	Version	Purpose
react-hook-form	7.88.0	Performant Form State Management
@hookform/resolvers	5.9.1	Validation Resolver Adapters
zod	4.6.5	Schema Validation (Type-safe)

File Processing & OCR

Technology	Version	Purpose
pdfjs-dist	5.4.296	PDF Parsing & Rendering (Mozilla)
pdf-parse	2.4.5	PDF Text Extraction

tesseract.js	7.0.0	OCR (Optical Character Recognition)
sharp	0.35.4	High-performance Image Processing (libvips)
xlsx (SheetJS)	0.20.3	Spreadsheet Parsing (Excel/CSV)
papaparse	5.7.0	CSV Parsing
qrcode	1.5.4	QR Code Generation

Payments, Integrations & Utilities

Technology	Version	Purpose
nodemailer	10.0.10	Email Delivery (SMTP/Transports)
nepali-date-converter	3.4.0	Nepali Calendar (Bikram Sambat) Support
date-fns	4.4.0	Modern Date/Time Utilities
date-fns-tz	3.2.0	Timezone-aware Date Handling
three	0.186.1	3D Graphics (Visualizations/Charts)

Background Workers & Processing

Component	Technology	Purpose
Worker Runner	tsx (TypeScript Execute)	Custom Worker System (src/workers/)
Job Handlers	TypeScript	Async Job Processing (Import, Prediction, Notifications)
Scheduling	Custom + Cron	Scheduled Tasks (Risk Engine, Reports, Cleanup)

Testing

Technology	Version	Purpose
vitest	5.0.1	Unit & Integration Testing (Vite-native)
playwright-core	1.63.0	E2E Browser Testing

Development & Build Tools

Technology	Version	Purpose
tsx	4.23.15	TypeScript Execution (Scripts, Workers, Seeds)
eslint	9.x	Linting (Flat Config)
eslint-config-next	16.3.6	Next.js ESLint Rules
dotenv	18.0.3	Environment Variable Management
PostCSS	via Tailwind 4	CSS Processing Pipeline

Key Architectural Highlights

- Multi-tenant SaaS Architecture — Complete data isolation per gym (tenant) via gymId on all models
- Feature Flags System — Three scopes (Global, Plan, Gym) with real-time toggles
- ML-powered Risk Engine — Member churn prediction with confidence scoring, factors, and revenue projections
- Automated Import Pipeline — CSV/Excel/PDF/Image → OCR → Column Mapping → Validation → Normalized Import
- Financial Ledger — Append-only double-entry style ledger for all money movements
- Marketplace Platform — Multi-vendor, commission rules, bookings, bundles, promotions
- Recovery Workflows — At-risk member outreach tracking (Call/SMS/WhatsApp/Email/In-person)
- Nepal Localization — NPR currency (paisa storage), Nepali calendar, Asia/Kathmandu timezone
- Server External Packages — Native/WASM modules (@node-rs/argon2, sharp, tesseract.js, pdfjs-dist, xlsx) configured for serverless

Deployment Target

Designed for **Vercel** (Next.js native platform) with serverless functions. Native/WASM modules configured via serverExternalPackages in next.config.ts. PostgreSQL database (managed: Vercel Postgres, Neon, Supabase, or self-hosted). Environment-based configuration via .env files.